Tecnología Digital VI

Transformers, LLMs, ViT

Licenciatura en Tecnología Digital - UTDT

GNMT

En 2016 Google revoluciona el problema de Machine Translation utilizando el mecanismo de atención buscando resolver el problema del *bottleneck* en la representación que buscar el encoder.

Utilizó una arquitectura de 8 capas de LSTM en el encoder y 8 en el decoder.

Más allá del breakthrough, se seguían utilizando LSTMs.

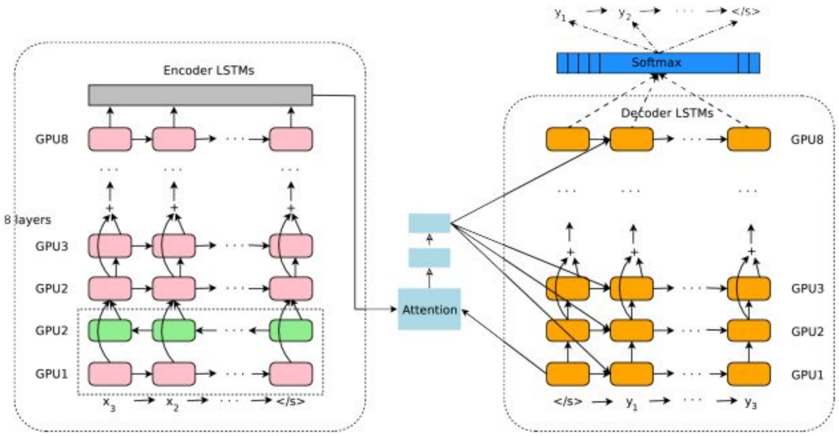


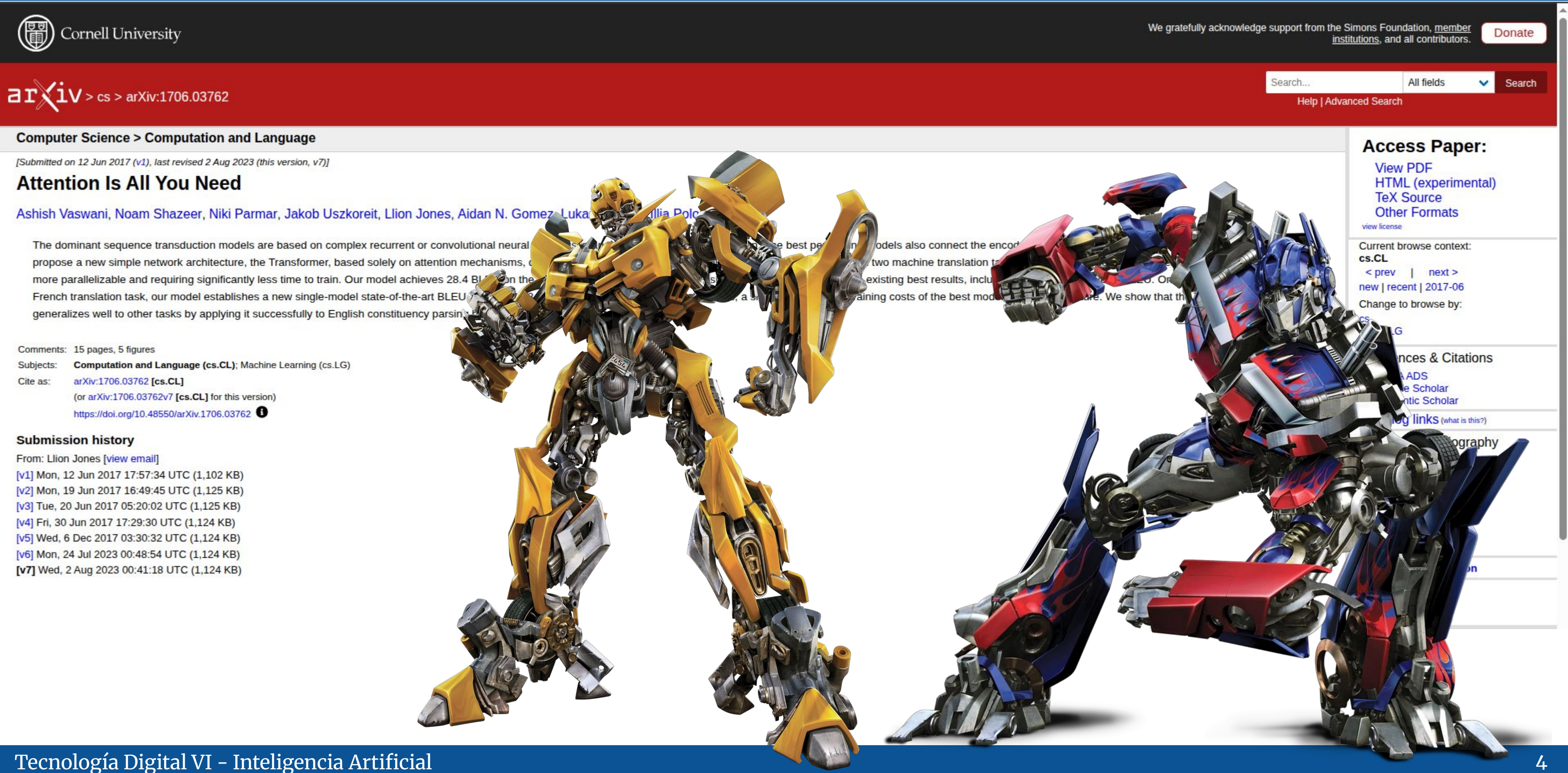
Figure 1: The model architecture of GNMT, Google's Neural Machine Translation system. On the left is the encoder network, on the right is the decoder network, in the middle is the attention module. The bottom encoder layer is bi-directional: the pink nodes gather information from left to right while the green nodes gather information from right to left. The other layers of the encoder are uni-directional. Residual connections start from the layer third from the bottom in the encoder and decoder. The model is partitioned into multiple GPUs to speed up training. In our setup, we have 8 encoder LSTM layers (1 bi-directional layer and 7 uni-directional layers), and 8 decoder layers. With this setting, one model replica is partitioned 8-ways and is placed on 8 different GPUs typically belonging to one host machine. During training, the bottom bi-directional encoder layers compute in parallel first. Once both finish, the uni-directional encoder layers can start computing, each on a separate GPU. To retain as much parallelism as possible during running the decoder layers, we use the bottom decoder layer output only for obtaining recurrent attention context, which is sent directly to all the remaining decoder layers. The softmax layer is also partitioned and placed on multiple GPUs. Depending on the output vocabulary size we either have them run on the same GPUs as the encoder and decoder networks, or have them run on a separate set of dedicated GPUs.

Problema de secuencialidad

El mecanismo de atención entre el encoder y el decoder permitía sortear la dificultad del cuello de botella en la representación del espacio latente, pero al utilizar RNNs todavía quedaba otro cuello de botella que era computacional. Las RNNs son inherentemente secuenciales en su cómputo, para conocer el estado en el paso 50 de la secuencia, es porque primero computé el estado en el paso 49.

Las CNNs no tenían este problema y por eso en esos años (2012-2016) pudieron aprovechar de manera mucho más eficiente los avances en procesamiento.

Transformers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Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Luka

The dominant sequence transduction models are based on complex recurrent or convolutional neural ... best pe... odels also connect the encod... propose a new simple network architecture, the Transformer, based solely on attention mechanisms, ... two machine translation ... more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BL... on the ... existing best results, inclu... French translation task, our model establishes a new single-model state-of-the-art BLEU ... aining costs of the best mod... We show that th... generalizes well to other tasks by applying it successfully to English constituency parsin...

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Attention Is All You Need

El trabajo de Google de 2017 es claro en su propuesta desde su título: no es necesario utilizar redes de procesamiento secuencial como las LSTMs.

El mismo mecanismo de atención que se utilizaba para relacionar al decoder con el encoder se puede usar como *ladrillo* fundamental.

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Self-Attention

En la clase pasada el mecanismo de atención que se utilizó relacionaba el decoder con el encoder. Es decir, la atención era cruzada entre la frase que estoy generando y la frase fuente que quiero traducir.

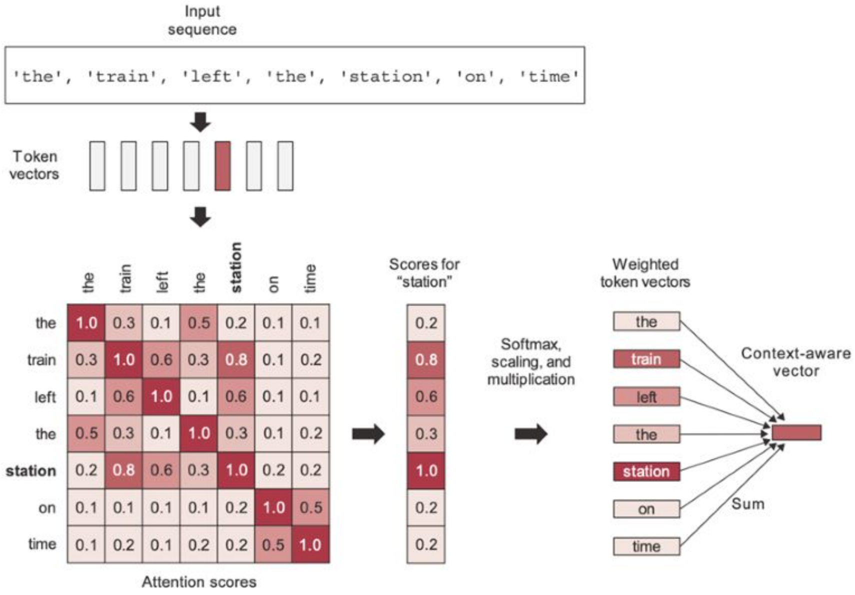
Si pretendemos reemplazar las LSTMs que forman al encoder o al decoder de manera independiente, necesitamos un mecanismo que pueda poner la atención sobre la misma frase que estoy procesando.

Ese mecanismo de atención se denomina *self-attention*. Las palabras de una oración se prestan atención a sí mismas para construir una representación interesante.

Self-Attention

La clase pasada vimos el mecanismo de atención aditivo de Bahdanau para calcular los scores de las palabras del encoder en la frase del decoder.

En Self-Attention cada palabra de una frase tendrá un vector de scores para esa misma frase. En vez de utilizar los scores *aditivos* que vimos, se utilizó el mecanismo de *scaled dot-product attention.*



Self-Attention vs Lookup Table

El mecanismo de *scaled dot-product attention* se suele relacionar con una *forma suavizada* de *lookup table*. ¿Qué significa esto?

En una *lookup table* (o simplemente un diccionario) se tiene una *query* q, se busca contra cuál de todas las *keys* k del diccionario corresponde, y luego se toma el *value* v correspondiente a ese k.

En el mecanismo de *scaled dot-product attention* también vamos a tener nuestra *query* q, *keys* k y *values* v, pero vamos a permitir que la q tenga una correspondencia parcial con varias de las k y vamos a tomar el promedio ponderado de las v en consecuencia.

Scaled Dot-Product Attention

El mecanismo hace el siguiente paso a paso:

1. Tenemos una secuencia de palabras w_i.
2. Convertimos las palabras w_i a su embedding x_i simplemente multiplicando por la matriz E del embedding.
3. Transformamos el embedding de cada palabra para conseguir su *query* q_i, su *key* k_i y su *value* v_i.
4. Calculamos las similaridades de todos los pares entre *queries* y *keys*.
5. Normalizamos las similaridades con una *softmax*.
6. Calculamos la suma ponderada de los *values*.

2) $$x_i = Ew_i$$

3) $$q_i = Qx_i \qquad k_i = Kx_i \qquad v_i = Vx_i$$

4) $$e_{ij} = \frac{q_i^\top k_j}{\sqrt{d_k}}$$

5) $$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{j'=1}^{n} \exp(e_{ij'})}$$

6) $$o_i = \sum_{j=1}^{n} \alpha_{ij} v_j$$

Scaled Dot-Product Attention

Al ver las cuentas que realiza el mecanismo surgen varias dudas posibles:

- ¿De dónde salen los valores de Q, K y V?
- ¿Por qué al calcular la similaridad hay un término dividiendo?
- ¿Por qué se aprenden Q y K por separado si después solamente usamos los e_{ij} de manera conjunta?
- ¿Cómo hacemos para que el mecanismo no ponga atención sobre palabras del *futuro*?
- ¿Cuál de las cuentas rompe la linealidad?
- Con estas cuentas perdimos la noción del orden de la palabras. ¿Cómo se soluciona esto?

Q,K,V

- ¿De dónde salen los valores de Q, K y V?

Es justamente lo que se aprende en el mecanismo. Vamos a comenzar con valores aleatorios y se irán ajustando para minimizar la *loss function* con el objetivo puntual que se tenga (por ejemplo, traducir bien).

Scaled Dot-Product

- ¿Por qué al calcular la similaridad hay un término dividiendo?

Cuando la dimensión de los vectores es grande, el producto interno que calculamos también se va haciendo grande.

La adición de la división por la raíz cuadrada de la cantidad de keys es lo que le da el nombre *scaled dot-product* al mecanismo y sirve simplemente para estabilizar las cuentas. La versión *dot-product* sin dicha división tiene la misma semántica en cuanto a lo que hace el mecanismo.

Low-Rank Factorization

- ¿Por qué se aprenden Q y K por separado si después solamente usamos los e_{ij} de manera conjunta?

Este es tal vez uno de los puntos más interesantes del mecanismo. La multiplicación de los q y k funciona como una aproximación de baja dimensión de los e_{ij}. Es una *low-rank factorization* de la matriz que realmente terminamos usando.

Esto por un lado es más eficiente computacionalmente, pero aparte obliga a que haya un cuello de botella en la representación y que cada valor de atención no pueda aprenderse de manera esparsa por sí solo.

Masked Attention

- ¿Cómo hacemos para que el mecanismo no ponga atención sobre palabras del *futuro*?

Si vamos a usar este mecanismo para reemplazar por ejemplo lo que sucede dentro del decoder, entonces las palabras se van generando de manera *online* y cuando tengo la palabra *t* no puedo calcular la atención sobre la palabra *t+1* porque todavía no la conozco.

Según la aplicación puntual en la que estemos usando el mecanismo se puede *enmascarar* parte del input. Esto significa fijar los *scores* de ciertas palabras en función de otras en un valor como *menos infinito* para indicar que no se debe poner absolutamente nada de atención ahí.

Linealidad

- ¿Cuál de las cuentas rompe la linealidad?

La aplicación de *softmax* rompe un poco la linealidad pero no es su objetivo. Todas las demás cuentas presentadas son composiciones lineales. Este mecanismo se va a utilizar luego en conjunto con formas clásicas de ruptura de linealidad como una capa *feed-forward* con funciones de activación

Positional Encoding

- Con estas cuentas perdimos la noción del orden de la palabras. ¿Cómo se soluciona esto?

Si no hacemos nada al respecto, esto vuelve a tener aspecto de *bag of words*.

La solución del paper original fue sumar un *positional encoding* a las palabras. Básicamente mediante una cuenta de senos y cosenos codificaron la posición de cada palabra en la frase para que esa información también entre a la red.

La noción de ingresar algún tipo de codificación de posición era muy útil pero la versión original *naive* se fue transformando fuertemente con el paso de los años.

Positional Encoding (Senos y Cosenos) -> Positional Encoding *aprendible* -> Relative Positional Encoding -> Rotary Positional Embedding y más

Transformer

Con todo lo mencionado, podemos definir entonces la arquitectura original del Transformer presentado en el paper.

Seguimos teniendo un *encoder* y un *decoder* pero ya no hay RNNs. El *encoder* y el *decoder* tienen sus propias capas de *self-attention*. Además hay una capa de *cross-attention* entre ellos.

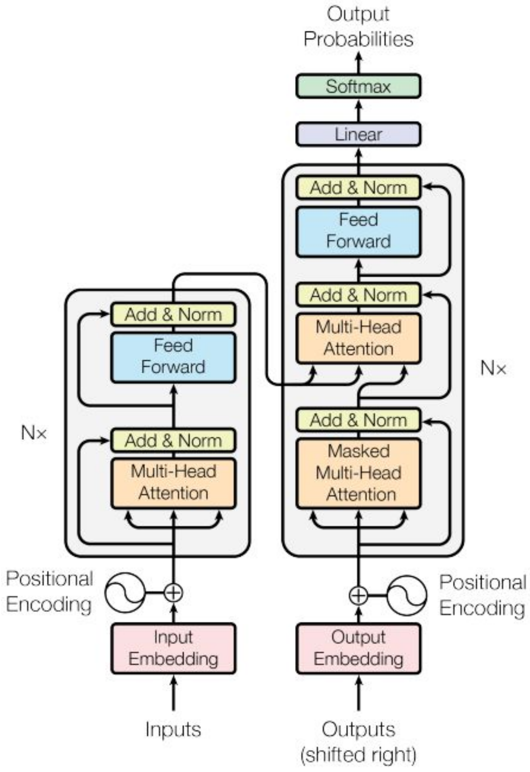


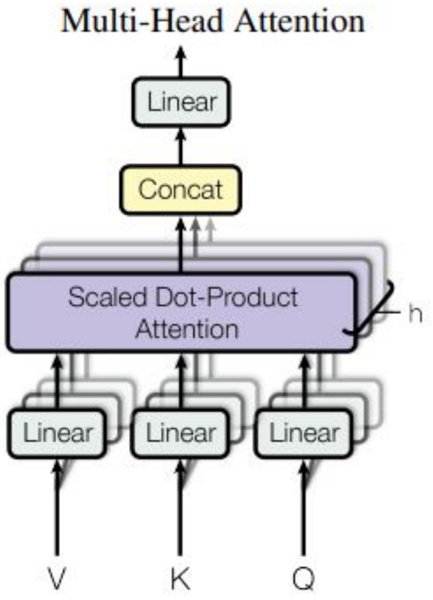
Figure 1: The Transformer - model architecture.

Multi-Head Attention

En las CNNs los parches de convoluciones tenían varios *canales* de salida porque sobre el mismo campo receptivo queríamos aprender a detectar diferentes cuestiones.

Sobre un mismo conjunto de palabras también queremos aprender diferentes representaciones. Tal vez una *cabeza* de atención esté aprendiendo a reconocer entidades puntuales (países, lugares, personas), mientras que otra *cabeza* de atención aprende a relacionar verbo y sujeto.

Además, las *multi-head attention* son computacionalmente muy eficientes.



Add & Norm

La capa de *add & norm* que se ve en la arquitectura en donde llegan los bloques de *multi-head attention* (y una skip connection desde lo previo) responde a dos operaciones:

- Layer Normalization: una capa de normalización (podemos pensarlo como reemplazo de BatchNorm) para acelerar la convergencia de la red.
- Residual Connections: las mismas conexiones residuales de la ResNet. Ayudan fuertemente en el aprendizaje gracias a como *viajan* los gradientes y el *landscape* que pasa a tener la *loss function*.

MNT

La arquitectura original se presentó como un *encoder* más un *decoder* con el objetivo de resolver el problema de traducción automática.

Ya en el trabajo original se consiguieron muy buenos resultados para dicho problema y con mucho menos entrenamiento que los modelos previos.

Adicionalmente, la arquitectura se podía reutilizar para otros problemas.

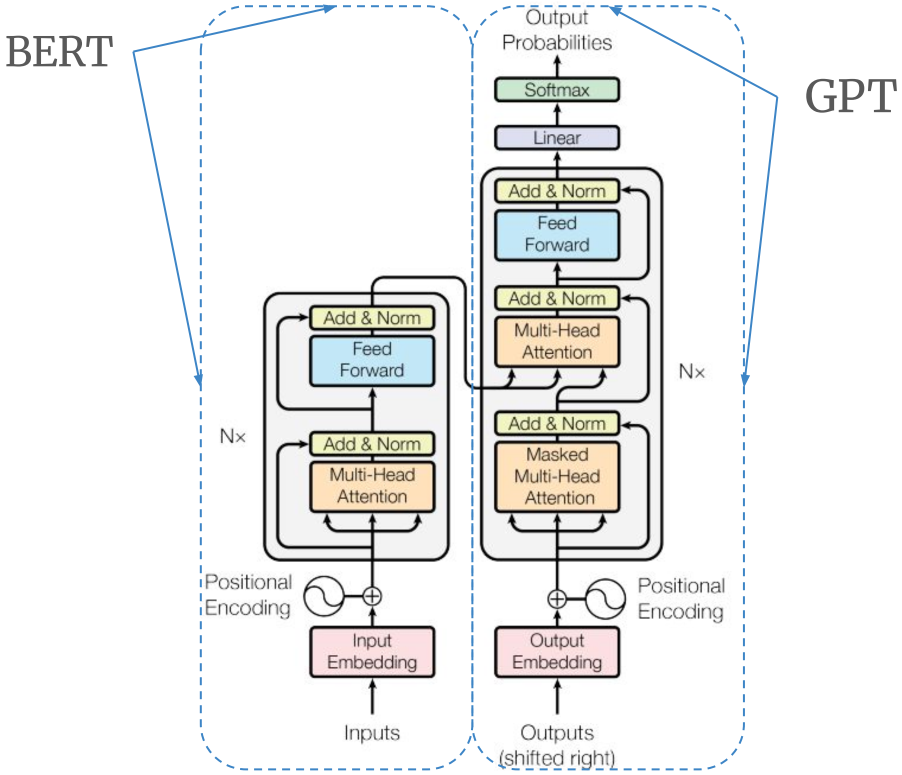
Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	**41.29**	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$\mathbf{3.3 \cdot 10^{18}}$	
Transformer (big)	**28.4**	**41.8**	$2.3 \cdot 10^{19}$	

BERT vs GPT

Si nos quedamos solo con el *encoder*, tenemos un gran codificador que puede tomar frases y llevarlas a un espacio de representación donde se podrán resolver varios problemas (clasificación, NER, Sentiment, etc).

Si nos quedamos solo con el *decoder*, tenemos un gran decodificador que puede generar frases.



BERT

En 2018 Google presenta BERT, un modelo que desde su nombre indica su arquitectura y cómo fue concebido.

Las representaciones obtenidas se pasaban por un proceso de *fine-tuning* para resolver (con excelentes resultados) diversos problemas de Procesamiento de Lenguaje Natural abiertos.

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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Abstract

We introduce a new language representation model called **BERT**, which stands for **B**idirectional **E**ncoder **R**epresentations from **T**ransformers. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering Test F1 to 93.2 (1.5 point absolute improvement) and SQuAD v2.0 Test F1 to 83.1 (5.1 point absolute improvement).

There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations.

We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches. The major limitation is that standard language models are unidirectional, and this limits the choice of architectures that can be used during pre-training. For example, in OpenAI GPT, the authors use a left-to-right architecture, where every token can only attend to previous tokens in the self-attention layers

GPT-1

También en 2018, y en realidad unos meses antes, OpenAI saca su paper sobre GPT-1.

Originalmente era algo bastante distinto a lo que terminó siendo. La *T* no era de *Transformer* (aunque sí era el modelo subyacente). La parte Generativa no era la protagonista, era una forma de resolver problemas de NLP.

Improving Language Understanding by Generative Pre-Training

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Abstract

Natural language understanding comprises a wide range of diverse tasks such as textual entailment, question answering, semantic similarity assessment, and document classification. Although large unlabeled text corpora are abundant, labeled data for learning these specific tasks is scarce, making it challenging for discriminatively trained models to perform adequately. We demonstrate that large gains on these tasks can be realized by *generative pre-training* of a language model on a diverse corpus of unlabeled text, followed by *discriminative fine-tuning* on each specific task. In contrast to previous approaches, we make use of task-aware input transformations during fine-tuning to achieve effective transfer while requiring minimal changes to the model architecture. We demonstrate the effectiveness of our approach on a wide range of benchmarks for natural language understanding. Our general task-agnostic model outperforms discriminatively trained models that use architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test), 5.7% on question answering (RACE), and 1.5% on textual entailment (MultiNLI).

GPT-2

GPT-1 perdía con claridad contra BERT en problemas NLP.

En el 2019 OpenAI presenta GPT-2 en donde el foco ya no era ganarle a BERT haciendo *fine-tuning* si no tener un modelo de lenguaje que pudiese resolver problemas de NLP sin prácticamente enseñarselo (zero-shot learning).

Language Models are Unsupervised Multitask Learners

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Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by the language model reach 55 F1 on the CoQA dataset - matching or exceeding the performance of 3 out of 4 baseline systems without using the 127,000+ training examples. The capacity of the language model is essential to the success of zero-shot task transfer and increasing it improves performance in a log-linear fashion across tasks. Our largest model, GPT-2, is a 1.5B parameter Transformer that achieves state of the art results on 7 out of 8 tested language modeling datasets in a zero-shot setting but still underfits WebText. Samples from the model reflect these improvements and contain coherent paragraphs of text. These findings suggest a promising path towards building language processing systems which learn to perform tasks from their naturally occurring demonstrations.

competent generalists. We would like to move towards more general systems which can perform many tasks – eventually without the need to manually create and label a training dataset for each one.

The dominant approach to creating ML systems is to collect a dataset of training examples demonstrating correct behavior for a desired task, train a system to imitate these behaviors, and then test its performance on independent and identically distributed (IID) held-out examples. This has served well to make progress on narrow experts. But the often erratic behavior of captioning models (Lake et al., 2017), reading comprehension systems (Jia & Liang, 2017), and image classifiers (Alcorn et al., 2018) on the diversity and variety of possible inputs highlights some of the shortcomings of this approach.

Our suspicion is that the prevalence of single task training on single domain datasets is a major contributor to the lack of generalization observed in current systems. Progress towards robust systems with current architectures is likely to require training and measuring performance on a wide range of domains and tasks. Recently, several benchmarks have been proposed such as GLUE (Wang et al., 2018) and decaNLP (McCann et al., 2018) to begin studying this.

Multitask learning (Caruana, 1997) is a promising framework for improving general performance. However, multitask training in NLP is still nascent. Recent work reports modest performance improvements (Yogatama et al.,

GPT-3

Para 2020, con GPT-3, el foco de OpenAI y sus GPTs ya era completamente abocado al Language Model en sí, y todo lo que podía resolver sin hacer nada más al respecto.

Language Models are Few-Shot Learners

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OpenAI

Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

InstructGPT

Un modelo de lenguaje no necesariamente es un *conversador*.

Un modelo de lenguaje bueno, por definición, genera secuencias de palabras que son correctas para el lenguaje subyacente, pero lo que genera puede no estar alineado con lo que el *usuario* le pidió.

InstructGPT llegó para resolver esa falta de alineación.

Training language models to follow instructions with human feedback

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OpenAI

Abstract

Making language models bigger does not inherently make them better at following a user's intent. For example, large language models can generate outputs that are untruthful, toxic, or simply not helpful to the user. In other words, these models are not *aligned* with their users. In this paper, we show an avenue for aligning language models with user intent on a wide range of tasks by fine-tuning with human feedback. Starting with a set of labeler-written prompts and prompts submitted through the OpenAI API, we collect a dataset of labeler demonstrations of the desired model behavior, which we use to fine-tune GPT-3 using supervised learning. We then collect a dataset of rankings of model outputs, which we use to further fine-tune this supervised model using reinforcement learning from human feedback. We call the resulting models *InstructGPT*. In human evaluations on our prompt distribution, outputs from the 1.3B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters. Moreover, InstructGPT models show improvements in truthfulness and reductions in toxic output generation while having minimal performance regressions on public NLP datasets. Even though InstructGPT still makes simple mistakes, our results show that fine-tuning with human feedback is a promising direction for aligning language models with human intent.

InstructGPT

En este ejemplo del paper se muestra como ante un *prompt* con una pregunta puntual GPT-3 completaba con las opciones *multiple choice* como si fuese un examen. Un output completamente razonable (e incluso esperado pensando los datos de entrenamiento) pero no *lindo* para una conversación.

Prompt:
What is the purpose of the list C in the code below?

```
def binomial_coefficient(n, r):
    C = [0 for i in range(r + 1)];
    C[0] = 1;
    for i in range(1, n + 1):
        j = min(i, r);
        while j > 0:
            C[j] += C[j - 1];
            j -= 1;
    return C[r]
```

GPT-3 175B completion:
A. to store the value of C[0]
B. to store the value of C[1]
C. to store the value of C[i]
D. to store the value of C[i - 1]

InstructGPT 175B completion:
The list C in this code is used to store the values of the binomial coefficient as the function iterates through the values of n and r. It is used to calculate the value of the binomial coefficient for a given value of n and r, and stores the result in the final return value of the function.

ChatGPT

En noviembre de 2022 OpenAI lanza ChatGPT. Un modelo *conversacional* en donde el *cerebro* era GPT-3 y su arquitectura de *Transformers* pero luego adaptado con diversas técnicas para ser mucho más ameno para conversar.

November 30, 2022 Product

Introducing ChatGPT

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We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests.

ChatGPT is a sibling model to InstructGPT, which is trained to follow an instruction in a prompt and provide a detailed response.

We are excited to introduce ChatGPT to get users' feedback and learn about its strengths and weaknesses. During the research preview, usage of ChatGPT is free. Try it now at chatgpt.com.

LLMs

Como vimos en el timeline de los GPTs, el “loro predictor” no alcanza. Para lograr un LLM, y en particular uno que a la gente le resulte útil para conversar, hace falta varias componentes adicionales.

A nivel *backbone* el modelo sigue siendo *el decoder de un transformer*, pero hay un montón de cuestiones interesantes alrededor para que sigan mejorando su funcionamiento día a día:

- Tokenizers.
- Data.
- SFT y RLHF.
- *Truquitos* de cómputo.
- y más...

ViT

En 2021 llegan los Vision Transformers. La idea del paper era parecida a *attention is all you need* mostrando que *no hacen falta* las CNNs.

Partían las imágenes en parches y luego aplicaban los mecanismos de atención entre los parches.

Consiguen muy buenos resultados pero a costa de tiempos de cómputo que incluso resultan prohibitivos para resoluciones altas.

La *batalla* sigue más viva que nunca y no hay un claro ganador para todo caso de uso (como sí sucedió en texto).

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AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

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ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.[1]

Multimodalidad 1.0

En el 2021 OpenAI publica su trabajo sobre CLIP (Constrastive Language-Image Pre-training).

Hasta el momento los embeddings de palabras y los embeddings de imágenes eran cuestiones separadas. CLIP busca relacionar esos embeddings haciendo que estén *alineados.*

Learning Transferable Visual Models From Natural Language Supervision

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Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at `https://github.com/OpenAI/CLIP`.

Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of "text-to-text" as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Over 20 years ago Mori et al. (1999) explored improving content based image retrieval by training a model to predict the nouns and adjectives in text documents paired with images. Quattoni et al. (2007) demonstrated it was possible to learn more data efficient image representations via manifold learning in the weight space of classifiers trained to predict words in captions associated with images. Srivastava & Salakhutdinov (2012) explored deep representation learning by training multimodal Deep Boltzmann Machines on top of low-level image and text tag features. Joulin et al. (2016) modernized this line of work and demonstrated that CNNs trained to predict words in image cap-

CLIP

CLIP busca entrenar un *encoder* para texto y un *encoder* para imágenes. La idea es que pueda predecir correctamente el par imagen-texto correcto, premiando el apareamiento esperado y penalizando los demás apareamientos posibles.

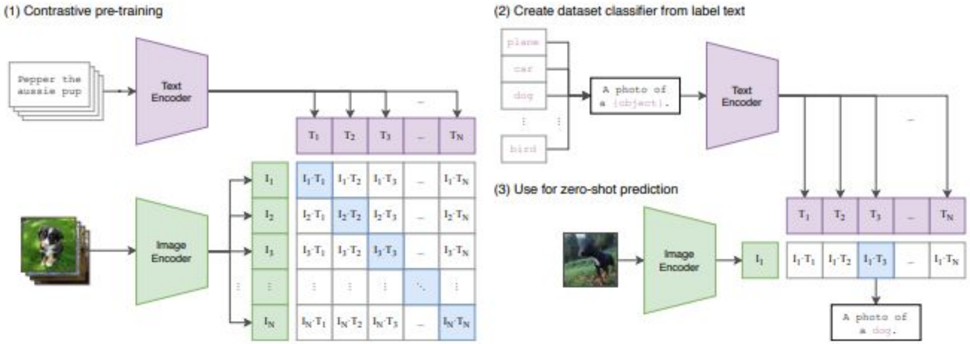

Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

Multimodalidad Nativa: Early-Fusion vs Late-Fusion

CLIP es un modelo *late-fusion* en el sentido de que toma el texto y las imágenes como dos cosas separadas y *las pega* al final.

Los modelos *early-fusion* plantearon la multimodalidad como algo *from start*, en donde directamente se entrene un modelo que puede esperar *todo* en el input.

Gemini: A Family of Highly Capable Multimodal Models

Gemini Team, Google[1]

This report introduces a new family of multimodal models, Gemini, that exhibit remarkable capabilities across image, audio, video, and text understanding. The Gemini family consists of Ultra, Pro, and Nano sizes, suitable for applications ranging from complex reasoning tasks to on-device memory-constrained use-cases. Evaluation on a broad range of benchmarks shows that our most-capable Gemini Ultra model advances the state of the art in 30 of 32 of these benchmarks — notably being the first model to achieve human-expert performance on the well-studied exam benchmark MMLU, and improving the state of the art in every one of the 20 multimodal benchmarks we examined. We believe that the new capabilities of the Gemini family in cross-modal reasoning and language understanding will enable a wide variety of use cases. We discuss our approach toward post-training and deploying Gemini models responsibly to users through services including Gemini, Gemini Advanced, Google AI Studio, and Cloud Vertex AI.

Chameleon: Mixed-Modal Early-Fusion Foundation Models

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We present Chameleon, a family of early-fusion token-based mixed-modal models capable of understanding and generating images and text in any arbitrary sequence. We outline a stable training approach from inception, an alignment recipe, and an architectural parameterization tailored for the early-fusion, token-based, mixed-modal setting. The models are evaluated on a comprehensive range of tasks, including visual question answering, image captioning, text generation, image generation, and long-form mixed modal generation. Chameleon demonstrates broad and general capabilities, including state-of-the-art performance in image captioning tasks, outperforms Llama-2 in text-only tasks while being competitive with models such as Mixtral 8x7B and Gemini-Pro, and performs non-trivial image generation, all in a single model. It also matches or exceeds the performance of much larger models, including Gemini Pro and GPT-4V, according to human judgments on a new long-form mixed-modal generation evaluation, where either the prompt or outputs contain mixed sequences of both images and text. Chameleon marks a significant step forward in a unified modeling of full multimodal documents.

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